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Course : MATH6198051 – Discrete Mathematics
Effective Period : September 2022

Logic
Session 1-3

Learning Objectives

On successful completion of this Course, students will be able to:



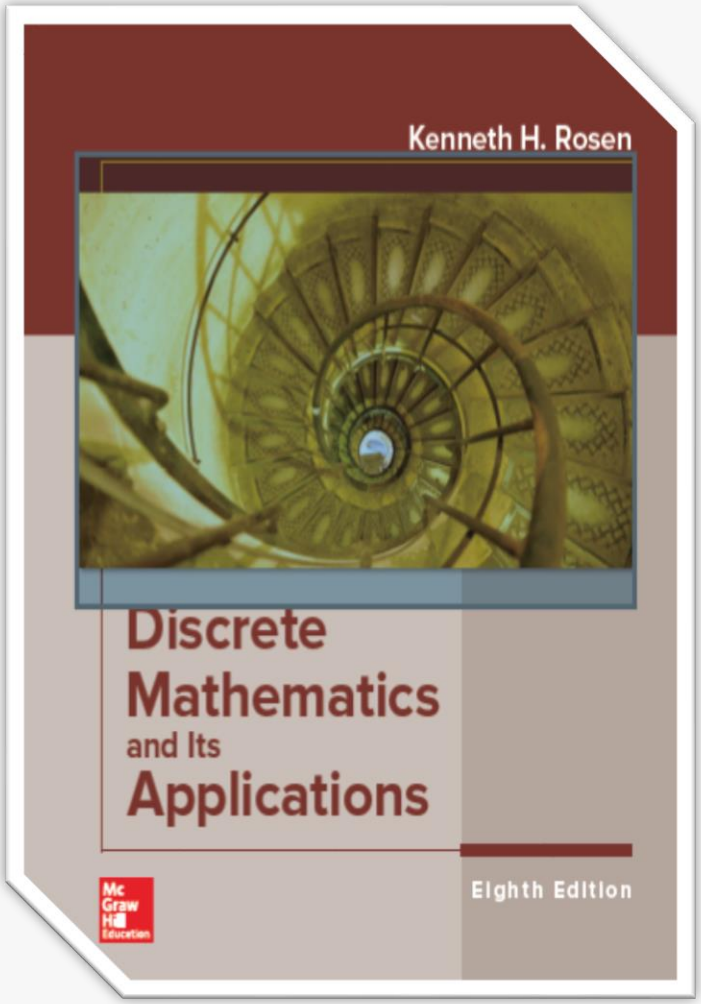
LO1:
Evaluate the logic of compound, quantified statements and method of proof related to discrete problems

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Acknowledgement



These slides have been adapted from :

Kenneth H. Rosen, " Discrete Mathematics and its Applications", 8th edition,2019, McGraw-Hill Education, New York, ISBN 978-1-259-67651-2

Chapter 1&12

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Sub Topics

1. Propositional Logic
2. Propositional Equivalences
3. Rules of Inference
4. Applications of Propositional Logic
5. Boolean Expressions and Boolean Functions
6. Logic Gates and Karnaugh Map

Propositional Logic

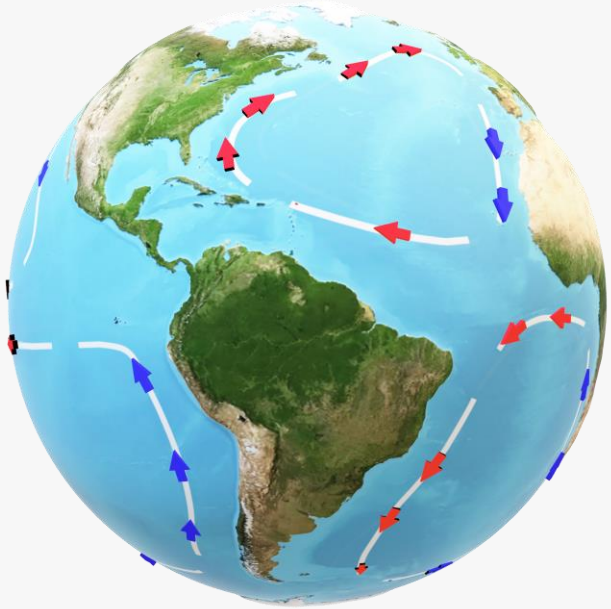
Propositions

A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

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Example :

1. Washington, D.C., is the capital of the United States of America. (T)
2. Toronto is the capital of Canada. (F)
3. 1 + 1 = 2. (T)
4. 2 + 2 = 3. (F)



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Propositions



Consider the following sentences :

1. What time is it?

   (not propositions , not declarative sentences)

2. Read this carefully.

   (not propositions, not declarative sentences)

3. $x + 1 = 2$.

   (not propositions - neither true nor false)

4. $x + y = z$.

   (not propositions - neither true nor false)

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Exclusive Or

Definition 4:

Let p and q be propositions. The ***exclusive or*** of p and q, denoted by $p \oplus q$ (or pXOR q), is

the proposition that is true when exactly one of p and q is true and is false otherwise.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Example : Let

p : "A student can have a salad with dinner" and

q : "A student can have soup with dinner,"

What is $p \oplus q$, the exclusive or of p and q?



Logical Operators

Solution :

"A student can have soup or a salad with dinner," (but not both)

Conditional Statements

Conditional Statements

Example : Let

p : "Maria learns discrete mathematics" and

q : "Maria will find a good job."



Then

$p \rightarrow q$: "If Maria learns discrete mathematics, then she will find a good job."

Other ways to express this conditional statement :

"Maria will find a good job when she learns discrete mathematics." **(*q* when *p*)**

"For Maria to get a good job, it is sufficient for her to learn discrete mathematics." **(*p* is sufficient for *q*)**

"Maria will find a good job unless she does not learn discrete mathematics."

(*q* unless $\neg p$)

Conditional Statements

There are three related conditional statements of $p \rightarrow q$:

1. **Converse** : $q \rightarrow p$
2. **Contrapositive** : $\neg q \rightarrow \neg p$
3. **Inverse** : $\neg p \rightarrow \neg q$

Example :

Find the contrapositive, the converse, and the inverse of the conditional statement

"The home team wins whenever it is raining."

Solution :

***p → q :* "If it is raining, then the home team wins."**

1. Contrapositive : "If the home team does not win, then it is not raining."
2. Converse : "If the home team wins, then it is raining."
3. Inverse : "If it is not raining, then the home team does not win."

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Biconditional Statements

Definition 6 :

Let p and q be propositions. The ***biconditional statement*** $p \leftrightarrow q$ is the proposition "p if and only if q." The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called *bi-implications*.

Biconditional Statement

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Ways to express this Biconditional statement:

"p is necessary and sufficient for q"
"if p then q, and conversely"
"p if q."
"p exactly when q."

Biconditional Statements



Example : Let

p : "You can take the flight," and

q : "You buy a ticket."

Then

$p \leftrightarrow q$: "You can take the flight if and only if you buy a ticket."

This statement is true if :

p and *q* are either both true or both false, that is, if you buy a ticket and can take the flight or if you do not buy a ticket and you cannot take the flight.

It is false when :

you do not buy a ticket, but you can take the flight (such as when you get a free trip) and when you buy a ticket but you cannot take the flight (such as when the airline bumps you).

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Precedence of Logical Operators

The precedence levels of the logical operators :

Operator	*Precedence*
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Example :

- $p \vee q \wedge r$ means $p \vee (q \wedge r)$
- $p \wedge q \vee r$ means $(p \wedge q) \vee r$
- $p \rightarrow q \vee r$ means $p \rightarrow (q \vee r)$
- $p \vee q \rightarrow r$ means $(p \vee q) \rightarrow r$

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Logic and Bit Operations

- Computers represent information using **bits**.
- A **bit** is a symbol with two possible values, namely,0 (zero) and 1 (one).
- The word bit comes from ***b*inary dig*it***, because zeros and ones are the digits used in binary representations of numbers.
- A bit can be used to represent a truth value, because there are two truth values, namely, ***true*** and ***false***. As is customarily done, we will use a 1 bit

  to represent true and a 0 bit to represent false.
- A variable is called a **Boolean variable** if its value is either true or false.

Truth Value	*Bit*
T	1
F	0

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Logic and Bit Operations

Definition 7 :

A *bit string* is a sequence of zero or more bits. The *length* of this string is the number of bits in the string.

Example : 101010011 is a bit string of length nine.

- ✓ The **bitwise *OR***, **bitwise *AND***, and **bitwise *XOR*** of two strings of the same length to be the strings that have as their bits the *OR*, *AND*, and *XOR* of the corresponding bits in the two strings, respectively.
- ✓ The symbols $\vee$, $\wedge$, and $\oplus$ to represent the bitwise *OR*, bitwise *AND*, and bitwise *XOR* operations

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Logic and Bit Operations

Example :

Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR* of the bit strings 01 1011 0110 and 11 0001 1101.

Solution:

01 1011 0110
11 0001 1101

11 1011 1111	**bitwise *OR***
01 0001 0100	**bitwise *AND***
10 1010 1011	**bitwise *XOR***



Propositional Equivalences

Compound proposition

Compound proposition : an expression formed from propositional variables using logical operators

Definisi 1.
A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called ***a tautology***. A compound proposition that is always false is called ***a contradiction***. A compound proposition that is neither a tautology nor a contradiction is called ***a contingency.***

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Example :

Tautology Contradiction

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logical Equivalences

Logically equivalent. : Compound propositions that have the same truth values in all possible cases.

Definisi 2.

The compound propositions p and q are called *logically equivalent* if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ ($p \Leftrightarrow$ q) denotes that p and q are logically equivalent.

Example : $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

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Logical Equivalences

$$p \to q \equiv \neg p \lor q$$

$$p \to q \equiv \neg q \to \neg p$$

$$p \lor q \equiv \neg p \to q$$

$$p \land q \equiv \neg(p \to \neg q)$$

$$\neg(p \to q) \equiv p \land \neg q$$

$$(p \to q) \land (p \to r) \equiv p \to (q \land r)$$

$$(p \to r) \land (q \to r) \equiv (p \lor q) \to r$$

$$(p \to q) \lor (p \to r) \equiv p \to (q \lor r)$$

$$(p \to r) \lor (q \to r) \equiv (p \land q) \to r$$

$$p \leftrightarrow q \equiv (p \to q) \land (q \to p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

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Logical Equivalences Identities

Equivalence	*Name*
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Constructing New Logical Equivalences

Example : Show that $\neg (p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

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Solution :

$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$ **by the second De Morgan law**

$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$ **by the first De Morgan law**

$\equiv \neg p \wedge (p \vee \neg q)$ **by the double negation law**

$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$ **by the second distributive law**

$\equiv \mathbf{F} \vee (\neg p \wedge \neg q)$ **because $\neg p \wedge p \equiv$ F**

$\equiv (\neg p \wedge \neg q) \vee \mathbf{F}$ **by the commutative law for disjunction**

$\equiv \neg p \wedge \neg q$ **by the identity law for F**

Consequently $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

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Rules of Inference

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Rules of Inference

- Proofs in mathematics are valid arguments that establish the truth of mathematical statements.
- By an **argument**, we mean a sequence of statements that end with a conclusion. By **valid**, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or **premises**, of the argument.
- An argument is **valid** if and only if it is impossible for all the premises to be true and the conclusion to be false.
- **Fallacies :** some common forms of incorrect reasoning which lead to invalid arguments.

Valid Arguments in Propositional Logic

Determine whether this is a valid argument !

"If you have a current password, then you can log onto the network."

"You have a current password."

Therefore,

"You can log onto the network."

Solution :

Let p : "You have a current password" and

q : "You can log onto the network."

Then, the argument has the form

$$p \rightarrow q$$
$$p$$
$$\therefore q$$

where $\therefore$ is the symbol that denotes "therefore."

Because the statement $((p \rightarrow q) \wedge p) \rightarrow q$ is a tautology so this argument is **valid.**

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Valid Arguments in Propositional Logic

Definition :

An ***argument*** in propositional logic is a sequence of propositions. All but the final proposition in the argument are called *premises* and the final proposition is called the ***conclusion***. An argument is ***valid*** if the truth of all its premises implies that the conclusion is true. An ***argument form*** in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is ***valid*** if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

Remark: From the definition of a valid argument form we see that the argument form with premises $p1$, $p2$,..., pn and conclusion q is valid exactly when $(p1 \wedge p2 \wedge \cdots \wedge pn) \rightarrow q$ is a **tautology.**

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Rules of Inference

Rule of Inference	*Tautology*	*Name*
p $p \rightarrow q$ $\therefore q$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\neg q$ $p \rightarrow q$ $\therefore \neg p$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism

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Rules of Inference

Rule of Inference	*Tautology*	*Name*
$\begin{array}{l} p \\ \hline \therefore\ p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore\ p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore\ p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore\ q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Using Rules of Inference to Build Arguments

Example :

Show that the premises :

"It is not sunny this afternoon and it is colder than yesterday,"

"We will go swimming only if it is sunny,"

"If we do not go swimming, then we will take a canoe trip," and

"If we take a canoe trip, then we will be home by sunset"

Click to add text

lead to the conclusion :

"We will be home by sunset."

Solution:

Let p : "It is sunny this afternoon,"

q : "It is colder than yesterday,"

r : "We will go swimming,"

s : "We will take a canoe trip," and

t : "We will be home by sunset."

$\therefore$ The conclusion is simply t.

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Using Rules of Inference to Build Arguments

Construct an argument :

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

So, each of the four hypotheses is true, the conclusion is also true

Resolution

- Many of computer programs make use of a rule of inference known as **resolution**. This rule of inference is based on the tautology,

  $((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$.

- The final disjunction in the resolution rule, $q \vee r$, is called the **resolvent.** When we let $q = r$ in this tautology, we obtain $(p \vee q) \wedge (\neg p \vee q) \rightarrow q$. Furthermore, when we let $r = \mathbf{F}$, we obtain $(p \vee q) \wedge (\neg p) \rightarrow q$ (because $q \vee \mathbf{F} \equiv q$), which is the tautology on which the rule of disjunctive syllogism is based.

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Example :

Use resolution to show that the hypotheses :

"Jasmine is skiing or it is not snowing" and
"It is snowing or Bart is playing hockey"
imply that
"Jasmine is skiing or Bart is playing hockey."

Resolution

Solution :

Let *p :* "It is snowing,"

q : "Jasmine is skiing," and

r : "Bart is playing hockey."

We can represent the hypotheses

"Jasmine is skiing or it is not snowing" **($\neg p \vee q$)**

"It is snowing or Bart is playing hockey" **($p \vee r$)**

Using resolution, the proposition :

$q \vee r$, ("Jasmine is skiing or Bart is playing hockey,") follows.

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Fallacies

1. Fallacy of affirming the conclusion.

The proposition $((p \rightarrow q) \wedge q) \rightarrow p$ is not a tautology, because it is false when p is false and q is true. However, there are many incorrect arguments that treat this as a tautology.

Example :

Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics.

Therefore, you did every problem in this book.

Solution :

Let p : "You did every problem in this book."

q : "You learned discrete mathematics."

$$\begin{array}{l} p \rightarrow q \\ q \\ \hline p \ ? \end{array}$$

This is **incorrect reasoning** called the **fallacy of affirming the conclusion**.

2. Fallacy of denying the hypothesis

The proposition $((p \rightarrow q) \wedge \neg p) \rightarrow \neg q$ is not a tautology, because it is false when p is false and q is true. Many incorrect arguments use this incorrectly as a rule of inference.

Example : Click to add text

Is it correct to assume that you did not learn discrete mathematics if you did not do every problem in the book, assuming that if you do every problem in this book, then you will learn discrete mathematics?

Solution :

$p \rightarrow q$
$\neg p$
$\neg q$?

Let p : "You did every problem in this book."

q : "You learned discrete mathematics."

It is possible that you learned discrete mathematics even if you did not do every problem in this book.

This is **incorrect reasoning** called the **Fallacy of denying the hypothesis**

Applications of Propositional Logic

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Applications of Propositional Logic

1. Translating English Sentences

Example :

How can this English sentence be translated into a logical expression?

"You can access the Internet from campus only if you are a computer science major or you are not a freshman."

Solution :

let

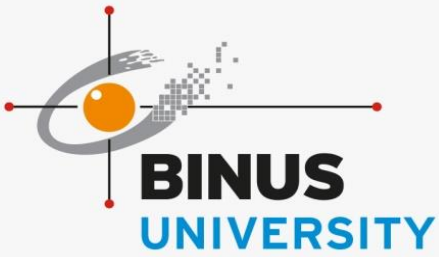
a : "You can access the Internet from campus"

c : "You are a computer science major"

f : "You are a freshman"

this sentence can be represented as

$$a \rightarrow (c \vee \neg f)$$



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Applications of Propositional Logic

2. **System Specifications**

Example:

Express the specification

"The automated reply cannot be sent when the file system is full"

using logical connectives.

Solution :

let p : "The automated reply can be sent" and

q : "The file system is full."

$\neg p$: "It is not the case that the automated reply canbe sent,"

Consequently, specification can be represented by the conditional statement

$$q \rightarrow \neg p$$

Applications of Propositional Logic

3. **Boolean Searches**

Example:

Web Page Searching : using Boolean searching to find Web pages about universities in New Mexico.

we can look for pages matching at search engines :

- NEW *AND* MEXICO *AND* UNIVERSITIES
- “NEW MEXICO” *AND* UNIVERSITIES.

Applications of Propositional Logic



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4. Logic Puzzles

Example :

As a reward for saving his daughter from pirates, the King has given you the opportunity to win a treasure hidden inside one of three trunks. The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk :

- Trunks 1 and 2 are each inscribed with the message "This trunk is empty," and
- Trunk 3 is inscribed with the message "The treasure is in Trunk 2."

The Queen, who never lies, tells you that **only one of these inscriptions is true, while the other two are wrong**. Which trunk should you select to win?

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Logic Puzzles

Solution :

Let p_i be the proposition that the treasure is in Trunk i, for $i = 1, 2, 3$. To translate into propositional logic the **Queen's statement** that exactly one of the inscriptions is true, we observe that the inscriptions on Trunk 1, Trunk 2, and Trunk 3, are $\neg p1$, $\neg p2$, and $p2$, respectively. So, her statement can be translated to

$$(\neg p1 \wedge \neg(\neg p2) \wedge \neg p2) \vee (\neg(\neg p1) \wedge \neg p2 \wedge \neg p2) \vee (\neg(\neg p1) \wedge \neg(\neg p2) \wedge p2)).$$
$$\equiv (p1 \wedge \neg p2) \vee (p1 \wedge p2).$$
$$\equiv p1 \wedge (\neg p2 \vee p2).$$
$$\equiv p1 \wedge T$$
$$\equiv p1$$

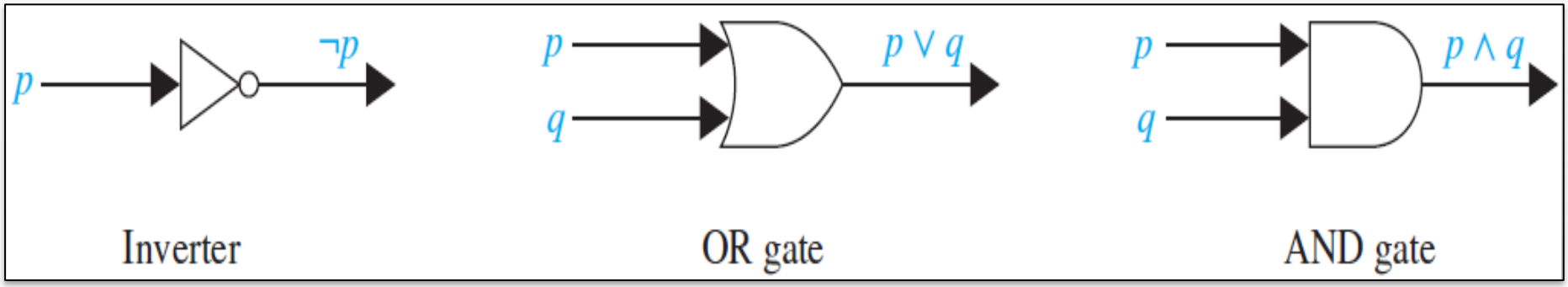
So the treasure is in Trunk 1 (that is, **$p1$ is true**), and $p2$ and $p3$ are false; and the **inscription** on Trunk 2 is the only true one.

Applications of Propositional Logic

5. Logic Circuits

- A **logic circuit** (or **digital circuit**) receives input signals $p1$, $p2$,..., pn, each a bit [either **0 (off)** or **1 (on)],** and produces output signals $s1$, $s2$,..., sn, each a bit.
- Complicated digital circuits can be constructed from three basic circuits, called **gates,** that is **NOT gate**, **OR gate**, and **AND gate**.

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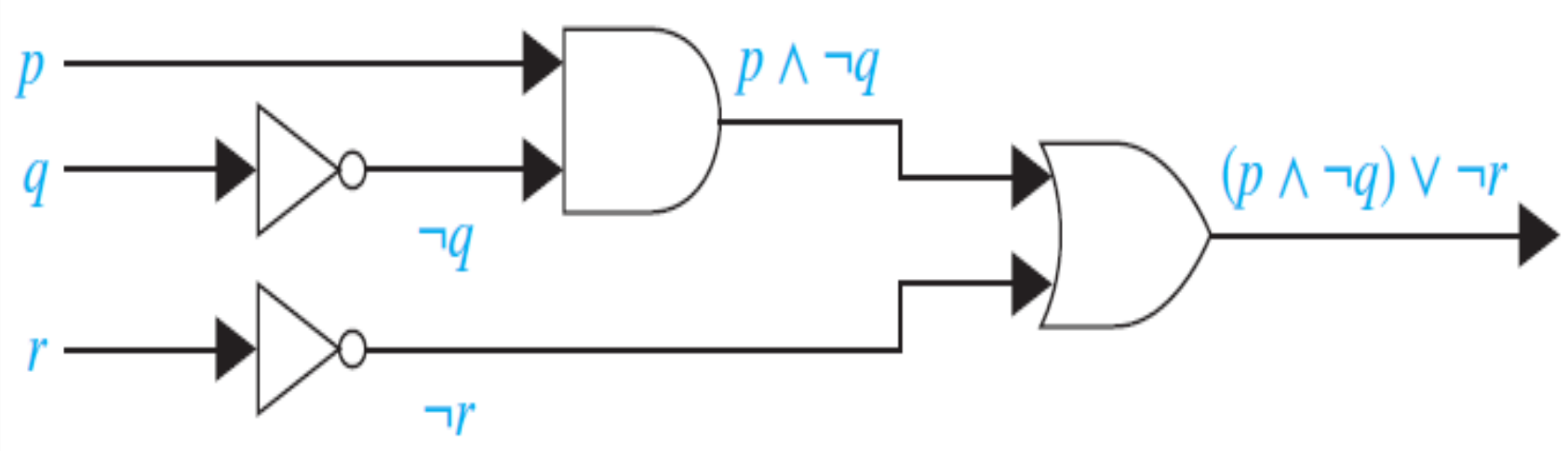


Logic Circuits

- Combinations of these three basic gates called **A combinatorial circuit.**

  t

  Example 1 : $(p \wedge \neg q) \vee \neg r$

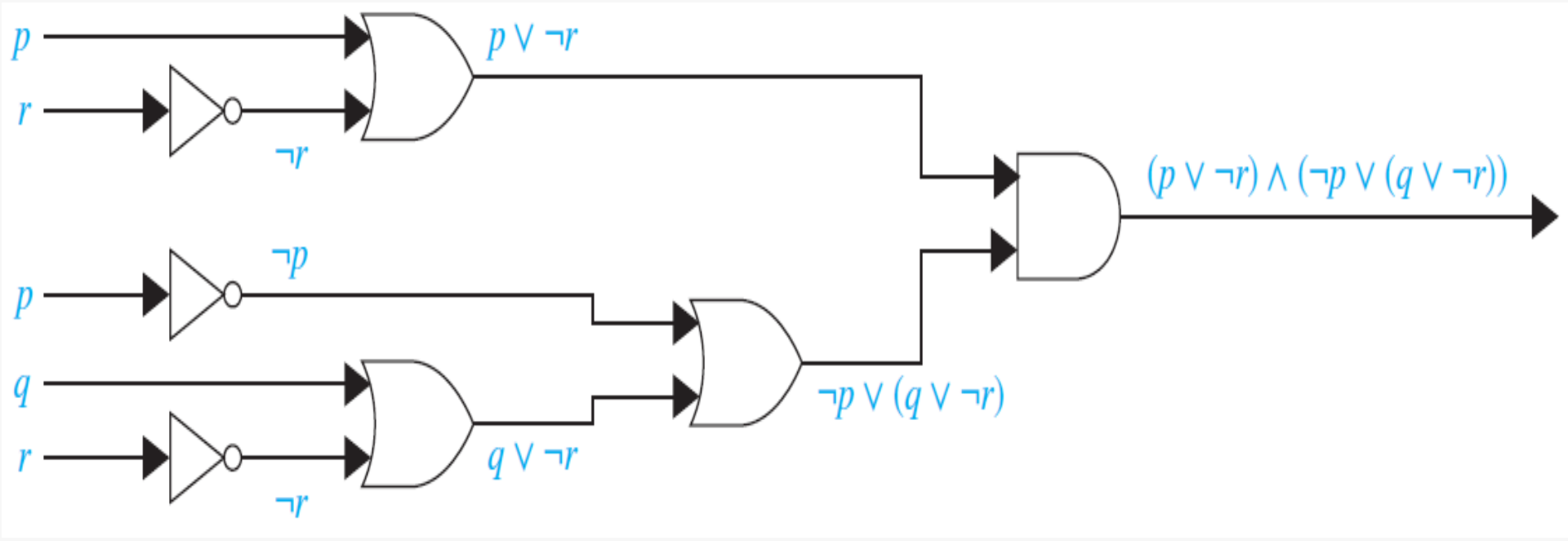


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A combinatorial circuit.

Example 2 :

$(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$



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Boolean Algebra

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Boolean expressions

Example

The values of the Boolean function represented by $F(x, y, z) = xy + \bar{z}$

x	y	z	xy	$\bar{z}$	$F(x, y, z) = xy + \bar{z}$
1	1	1	1	0	1
1	1	0	1	1	1
1	0	1	0	0	0
1	0	0	0	1	1
0	1	1	0	0	0
0	1	0	0	1	1
0	0	1	0	0	0
0	0	0	0	1	1

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Identities of Boolean Algebra

Identity	*Name*
$\bar{\bar{x}} = x$	Law of the double complement
$x + x = x$ $x \cdot x = x$	Idempotent laws
$x + 0 = x$ $x \cdot 1 = x$	Identity laws
$x + 1 = 1$ $x \cdot 0 = 0$	Domination laws
$x + y = y + x$ $xy = yx$	Commutative laws

Identity	*Name*
$x + (y + z) = (x + y) + z$ $x(yz) = (xy)z$	Associative laws
$x + yz = (x + y)(x + z)$ $x(y + z) = xy + xz$	Distributive laws
$\overline{(xy)} = \bar{x} + \bar{y}$ $\overline{(x + y)} = \bar{x}\,\bar{y}$	De Morgan's laws
$x + xy = x$ $x(x + y) = x$	Absorption laws
$x + \bar{x} = 1$	Unit property
$x\bar{x} = 0$	Zero property

Identities of Boolean Algebra

Example 1 :

Show that the distributive law $x(y + z) = xy + xz$ is valid.

Solution :

The identity holds because the last two columns of the table agree.

x	y	z	$y+z$	xy	xz	$x(y+z)$	$xy+xz$
1	1	1	1	1	1	1	1
1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

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Identities of Boolean Algebra

Example 2 :

Prove the **absorption law** $x(x + y) = x$ using the other identities of Boolean algebra

Solution:

$x(x + y) = (x + 0)(x + y)$	Identity law for the Boolean sum
$= x + 0 \cdot y$	Distributive law of the Boolean sum over the Boolean product
$= x + y \cdot 0$	Commutative law for the Boolean product
$= x + 0$	Domination law for the Boolean product
$= x$	Identity law for the Boolean sum.

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Representing Boolean Functions

1. Sum-of-Products Expansions / disjunctive normal form

Example 1 :

Find Boolean expressions that represent the functions $F(x, y, z)$ and $G(x, y, z)$, which are given in Table below

x	y	z	F	G
1	1	1	0	0
1	1	0	0	1
1	0	1	1	0
1	0	0	0	0
0	1	1	0	0
0	1	0	0	1
0	0	1	0	0
0	0	0	0	0

Solution :

$F(x, y,z) = x\bar{y}z$

$G(x, y, z) = xy\bar{z} + \bar{x}y\bar{z}$

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Representing Boolean Functions

1. Sum-of-Products Expansions / disjunctive normal form

Example 2 :

Find the sum-of-products expansion for the function $F(x, y, z) = (x + y)\,\bar{z}$

Solution :

a. Using Boolean identities

$F(x, y, z) = (x + y)\,\bar{z}$

$= x\,\bar{z} + y\,\bar{z}$ — Distributive law

$= x1\,\bar{z} + 1y\,\bar{z}$ — Identity law

$= x(y + \bar{y})\,\bar{z} + (x + \bar{x})y\,\bar{z}$ — Unit property

$= \mathbf{xy\,\bar{z}} + x\,\bar{y}\,\bar{z} + \mathbf{xy\,\bar{z}} + \bar{x}y\,\bar{z}$ — Distributive law

$= xy\,\bar{z} + x\,\bar{y}\,\bar{z} + \bar{x}y\,\bar{z}$ — Idempotent law

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Representing Boolean Functions

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Solution :

b. *B*y determining the values of *F* for all possible values of the variables *x*, *y*, and *z*

x	y	z	$x + y$	$\bar{z}$	$(x + y)\bar{z}$
1	1	1	1	0	0
1	1	0	1	1	1
1	0	1	1	0	0
1	0	0	1	1	1
0	1	1	1	0	0
0	1	0	1	1	1
0	0	1	0	0	0
0	0	0	0	1	0

This gives :

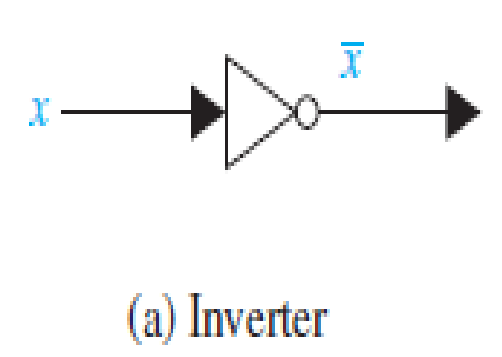
$F(x, y, z) = xy\,\bar{z} + x\,\bar{y}\,\bar{z} + \bar{x}\,y\,\bar{z}$

Representing Boolean Functions

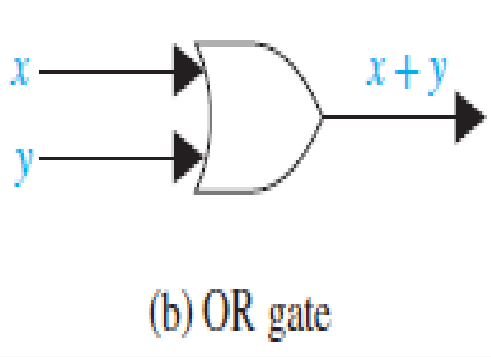
Representing Boolean Functions

- The basic elements of circuits are called **gates**
- The circuits that give output that depends only on the input, and not on the currentstate of the circuit. In other words, these circuits have no memory capabilities. Such circuits are called **combinational circuits** or **gating networks**.

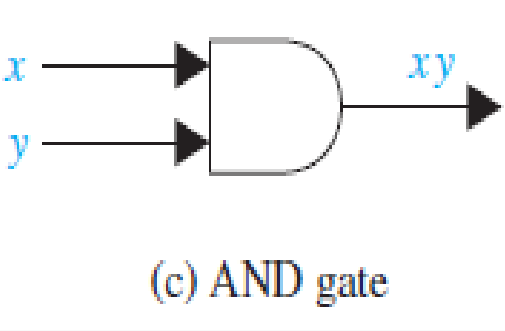
- **Basic types of gates.**



(a) Inverter



(b) OR gate



(c) AND gate

Combinations of Gates

Example

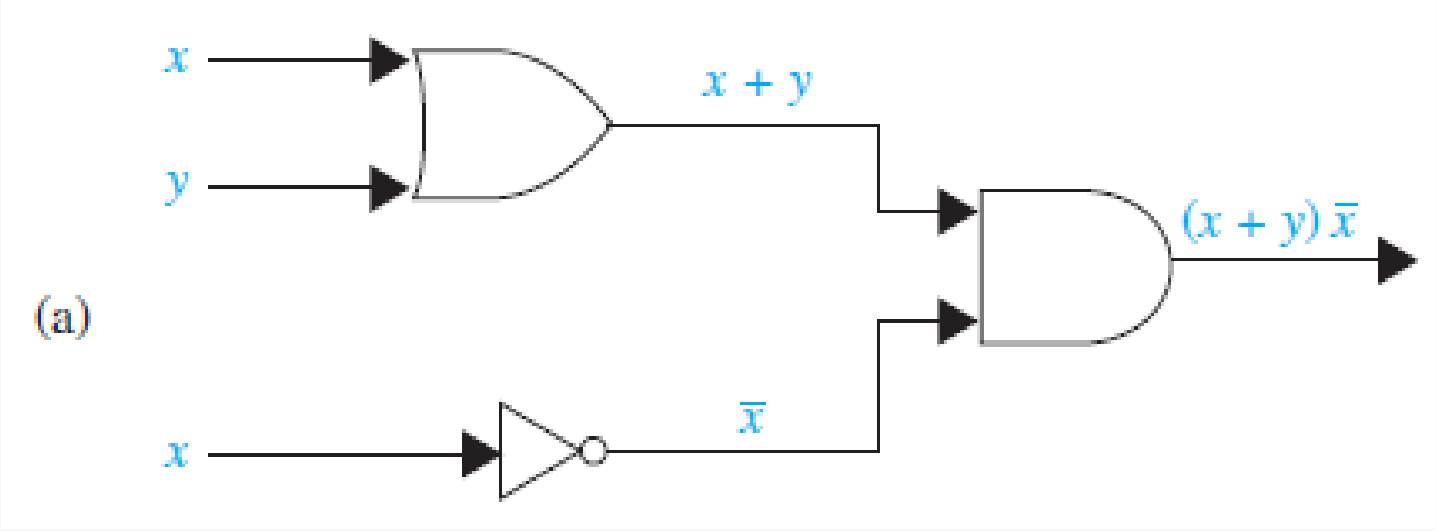
Construct circuits that produce the following outputs:

(a) $(x + y)\,\bar{x}$

(b) $\bar{x}\,\overline{(y + \bar{z})}$, and

(c) $(x + y + z)(\bar{x}\,\bar{y}\,\bar{z})$.

Solution :

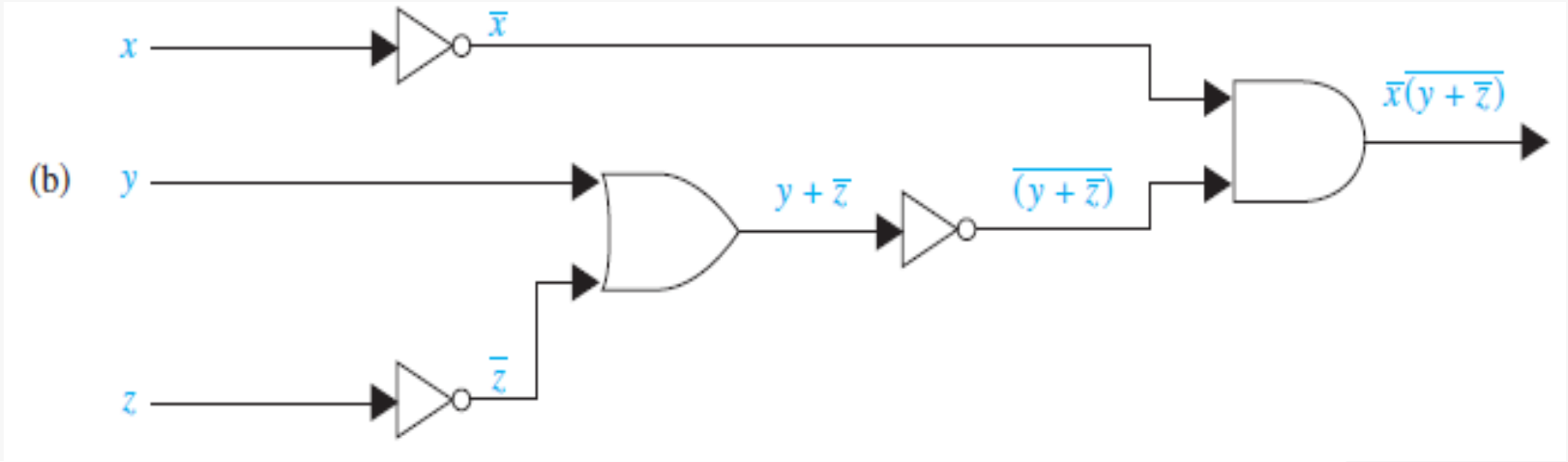


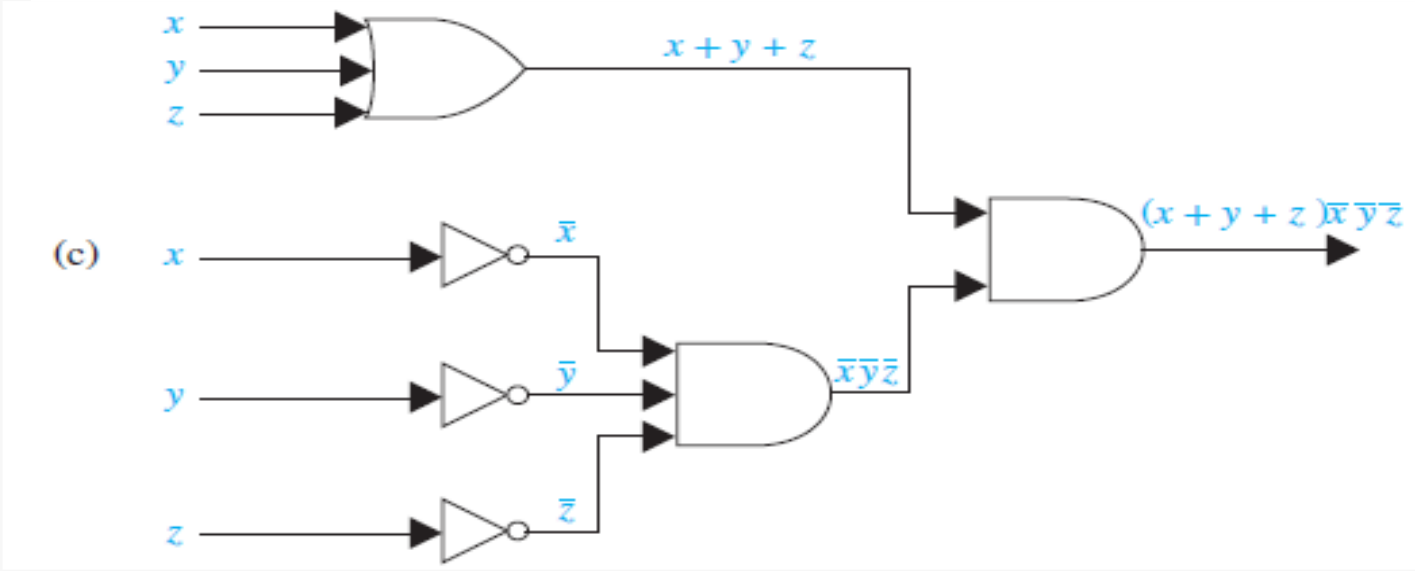
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Combinations of Gates





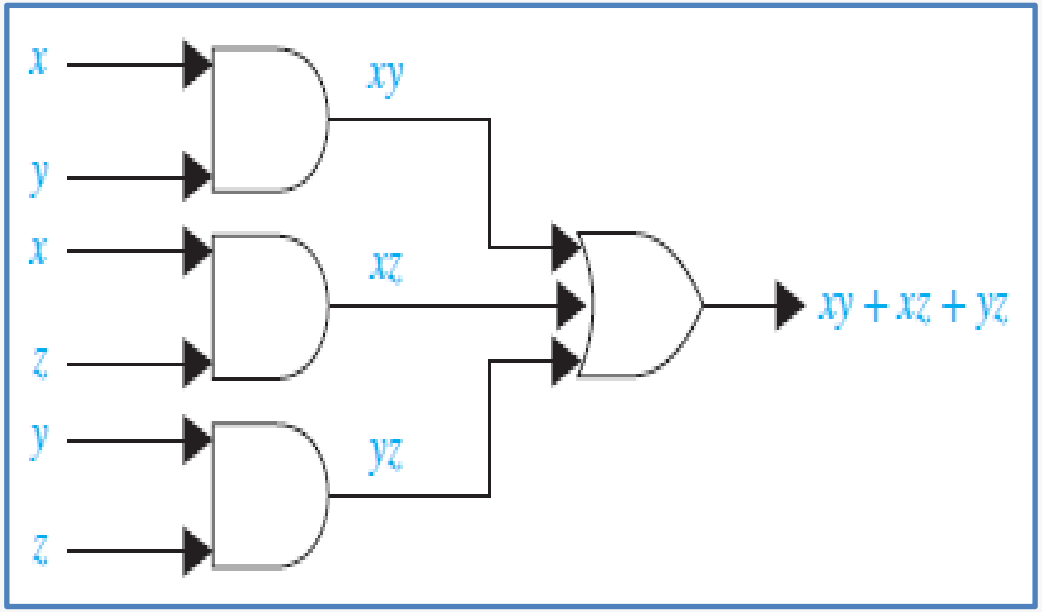
Examples of Circuits

Example 1 . Majority Voting

A committee of three individuals decides issues for an organization. Each individual votes either yes or no for each proposal that arises. A proposal is passed if it receives at least two yes votes. Design a circuit that determines whether a proposal passes.

Solution :

Let
x = 1 if the first individual votes yes,
y = 1 if the second individual votes yes,
z = 1 if the third individual votes yes,



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Examples of Circuits

Example 2 . A light controlled by two switches

Sometimes light fixtures are controlled by more than one switch. Circuits need to be designed so that flipping any one of the switches for the fixture turns the light on when it is off and turns the light off when it is on. Design circuits that accomplish this when there are two switches.

Solution :

Let

x = 1 when the first switch is closed

y = 1 when the second switch is closed

$F(x, y)$ = 1 when the light is on

$F(x, y)$ = 0 when the light is off

$F(1, 1)$ = 1 (Decide that the light will be on when both switches are closed)

$F(1, 0) = F(0, 1)$ = 0 (When one of the two switches is opened, the light goes off,

$F(0, 0)$ = 1 (When the other switch is also opened, the light goes on)

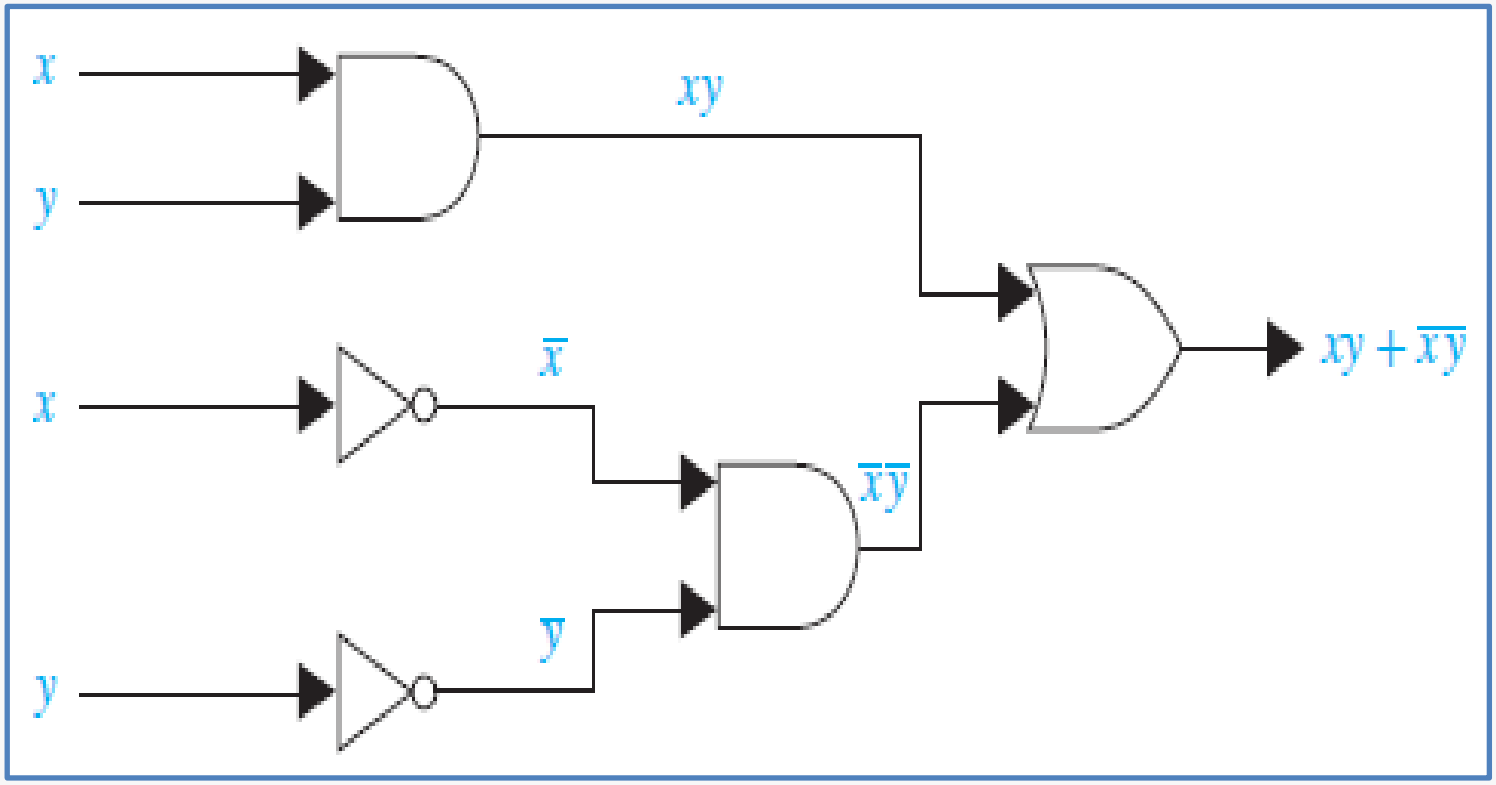
$$F(x, y) = xy + \bar{x}\,\bar{y}$$

x	y	$F(x, y)$
1	1	1
1	0	0
0	1	0
0	0	1

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A Circuit light controlled by two switches

Solution :



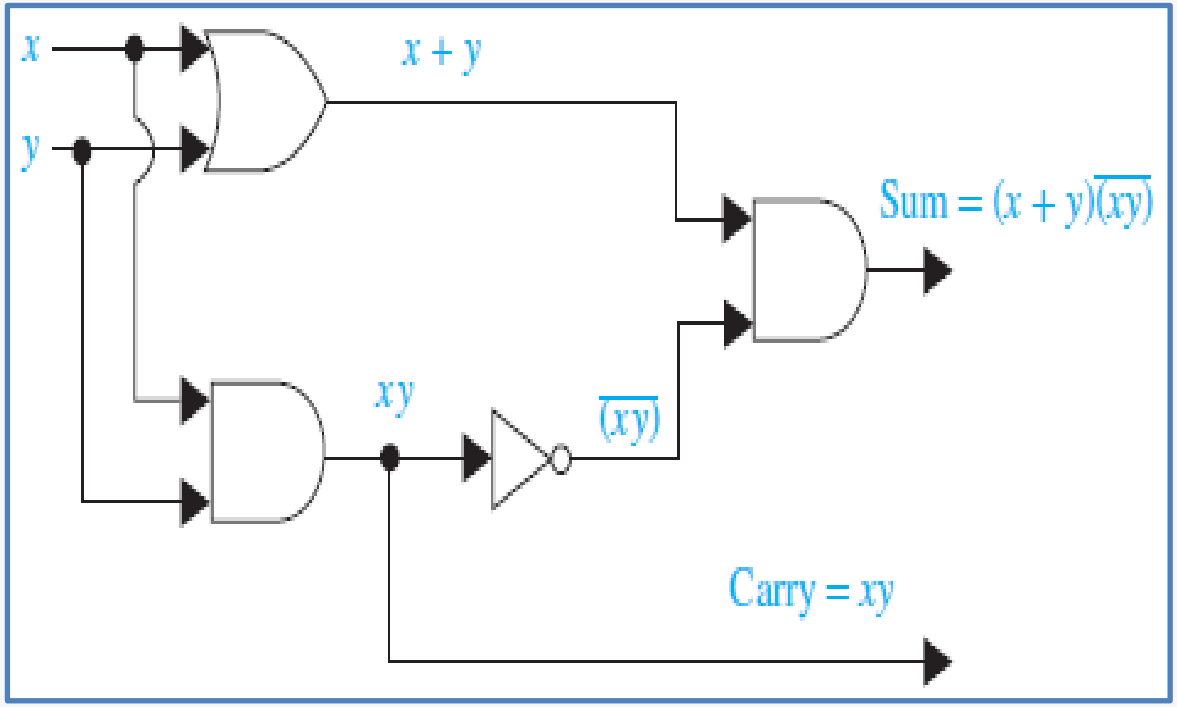
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Adders

The circuit that has two output is called the **half adder**, namely, *s* and *c*, where *s* is the sum bit and *c* is the carry bit .

Example

Input		*Output*	
x	***y***	***s***	***c***
1	1	0	1
1	0	1	0
0	1	1	0
0	0	0	0



Output for the half adder :

$$c = xy$$

$$s = x\bar{y} + \bar{x}y = (x + y)\overline{(xy)}.$$

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Karnaugh Map

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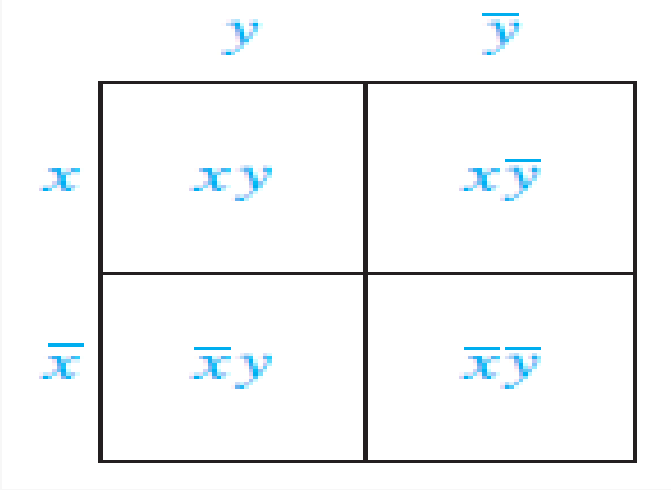
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Karnaugh Map

- A **Karnaugh map** or **K-map**, for finding terms to combine for Boolean functions involving a relatively small number of variables
- The method we will describe was introduced by Maurice Karnaugh in 1953. His method is based on earlier work by E.W. Veitch.
- K-maps give us a visual method for simplifying sum-of-products expansions; they are not suited for mechanizing this process
- A Karnaugh map has zero and one entries at different positions. Each position in a grid corresponds to a truth table entry.
- Given a Boolean function described by a truth table or logic function, then:
  - Draw the Karnaugh Map for the function.
  - Use the information from a Karnaugh Map to determine the smallest sum-of-products function.

Karnaugh Map in Two Variables

- ✓ There are four possible minterms in the sum-of-products expansion of a Boolean function in the two variables *x* and *y*.
- ✓ A K-map for a Boolean function in these two variables consists of four cells, where a 1 is placed in the cell representing a minterm if this minterm is present in the expansion. Cells are said to be **adjacent** if the minterms that they represent differ in exactly one literal.
- ✓ For instance, the cell representing $\bar{x}y$ is adjacent to the cell representing xy and $\bar{x}\bar{y}$.



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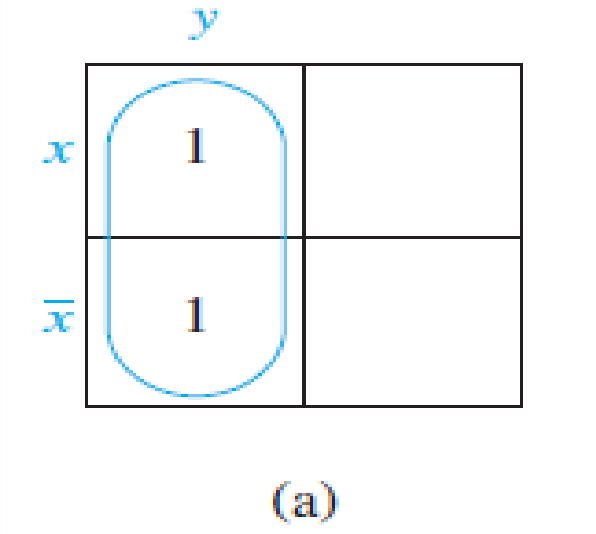
Karnaugh Map in Two Variables

Example :

Find the K-maps for Boolean expression E that equal to :

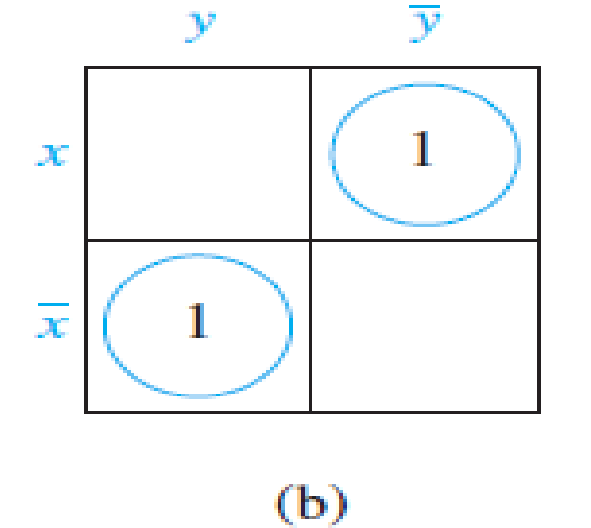
$$\text{(a) } xy + \bar{x}y, \text{ (b) } x\bar{y} + \bar{x}y, \text{ and (c) } x\bar{y} + \bar{x}y + \bar{x}\,\bar{y}.$$

Solution :



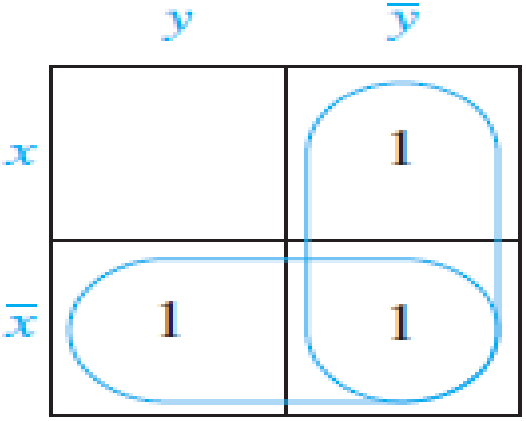
(a)

E = y



(b)

E = $x\bar{y} + \bar{x}y$,



(c)

E = $\bar{x} + \bar{y}$.

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Karnaugh Map in three variables

Let $\alpha(x,y,z)$ be a sum of product Boolean expression such that minterm consists of three variables x,y,z. To define the K-map for $\alpha(x,y,z)$, we consider the following table

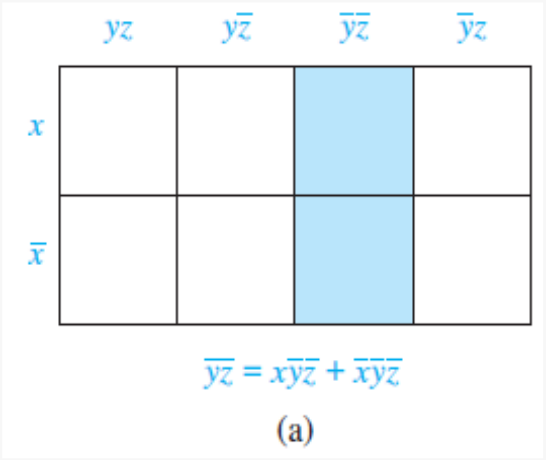
	yz	$y\bar{z}$	$\bar{y}\bar{z}$	$\bar{y}z$
x	xyz	$xy\bar{z}$	$x\bar{y}\bar{z}$	$x\bar{y}z$
$\bar{x}$	$\bar{x}yz$	$\bar{x}y\bar{z}$	$\bar{x}\bar{y}\bar{z}$	$\bar{x}\bar{y}z$

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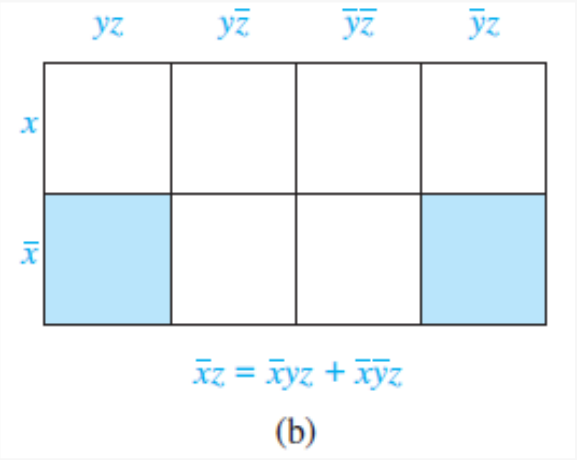
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Blocks in K-maps in three variables.



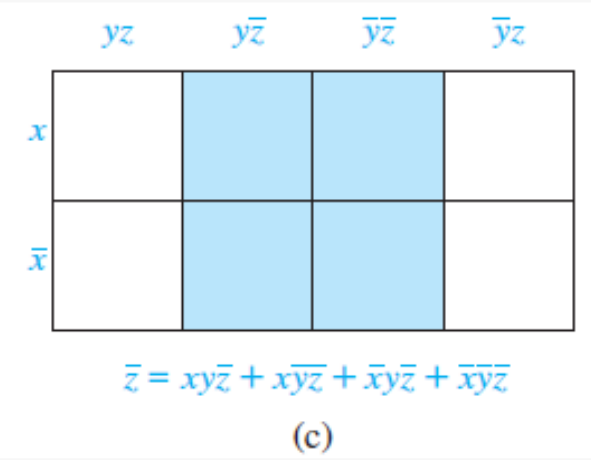
$\bar{y}\bar{z} = x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z}$

(a)



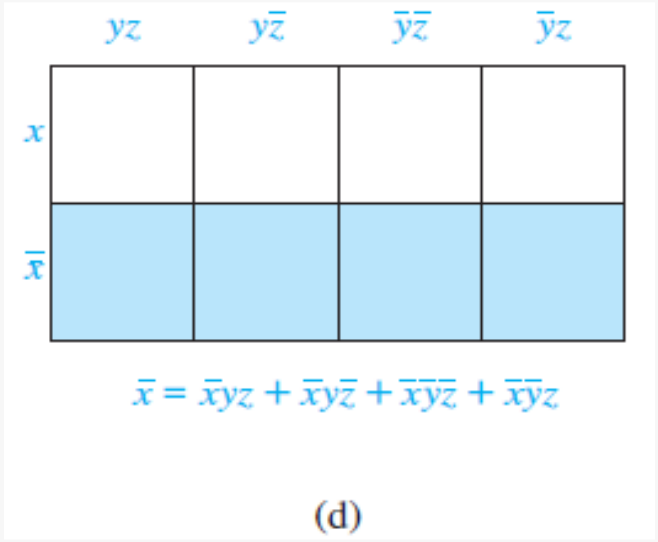
$\bar{x}z = \bar{x}yz + \bar{x}\bar{y}z$

(b)



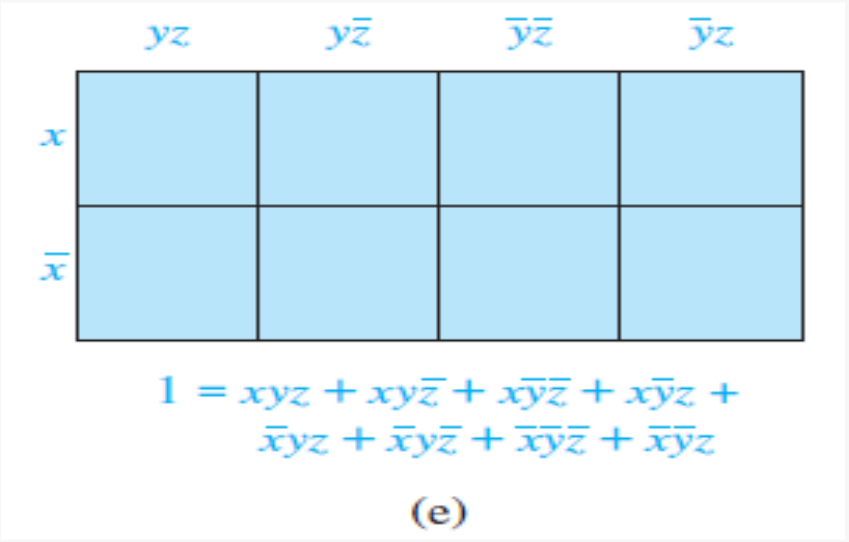
$\bar{z} = xy\bar{z} + x\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z}$

(c)



$\bar{x} = \bar{x}yz + \bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z$

(d)



$1 = xyz + xy\bar{z} + x\bar{y}\bar{z} + x\bar{y}z + \bar{x}yz + \bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z$

(e)

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Karnaugh Map in three variables

Example :

Use K-maps to minimize these sum-of-products expansions

(a) $xy\bar{z} + x\bar{y}\,\bar{z} + \bar{x}yz + \bar{x}\,\bar{y}\,\bar{z}$

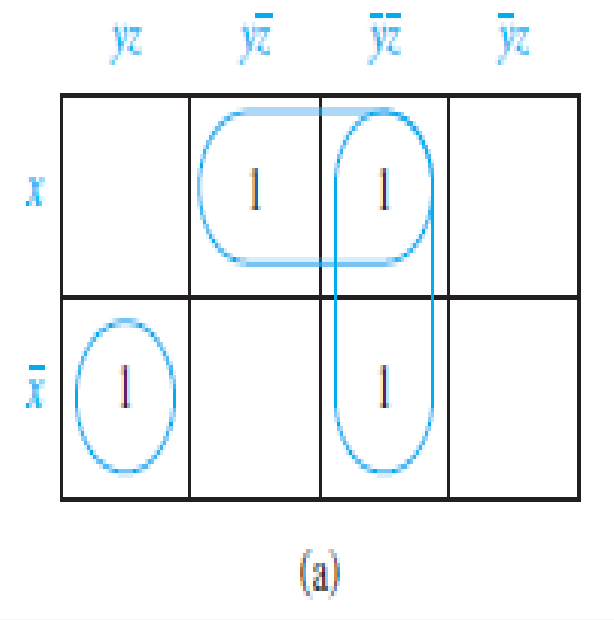
(b) $xy\bar{z} + x\bar{y}\,\bar{z} + \bar{x}yz + \bar{x}\,\bar{y}z + \bar{x}\,\bar{y}\,\bar{z}$

(c) $xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\,\bar{z} + \bar{x}yz + \bar{x}\,\bar{y}z + \bar{x}\,\bar{y}\,\bar{z}$

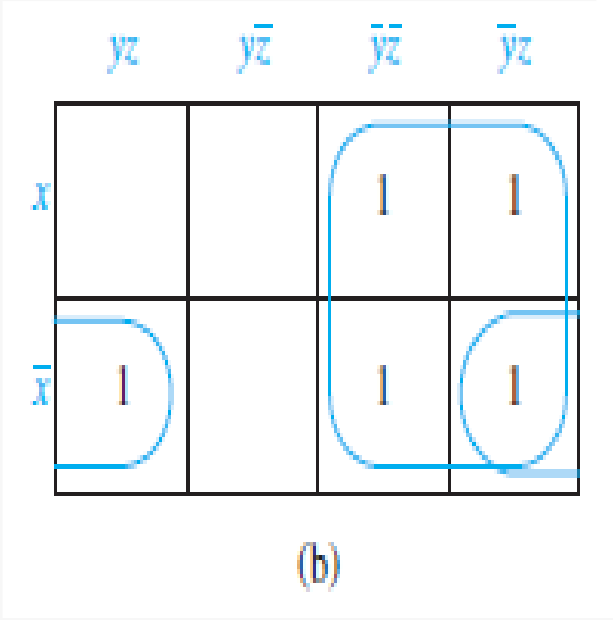
Karnaugh Map in three variables

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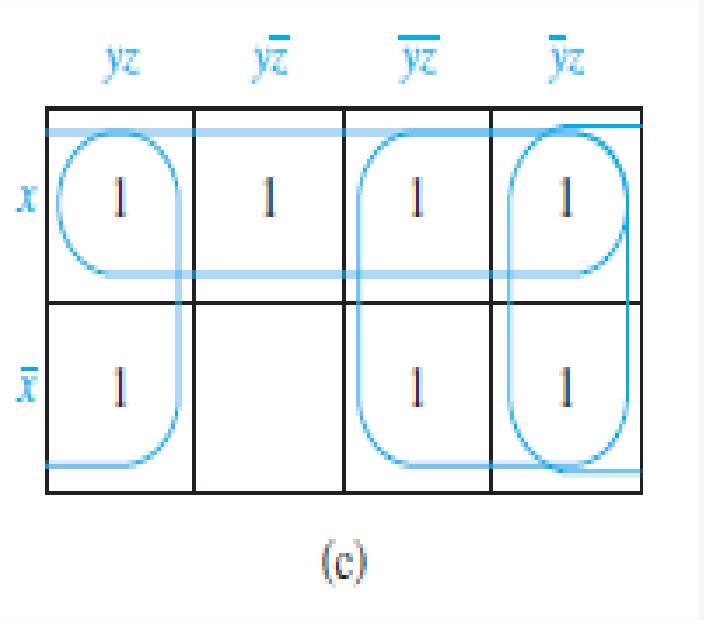
Solution :



$$x\bar{z} + \bar{y}\bar{z} + \bar{x}yz$$



$$\bar{y} + \bar{x}z$$



$$x + \bar{y} + z$$

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A K-map in four variables

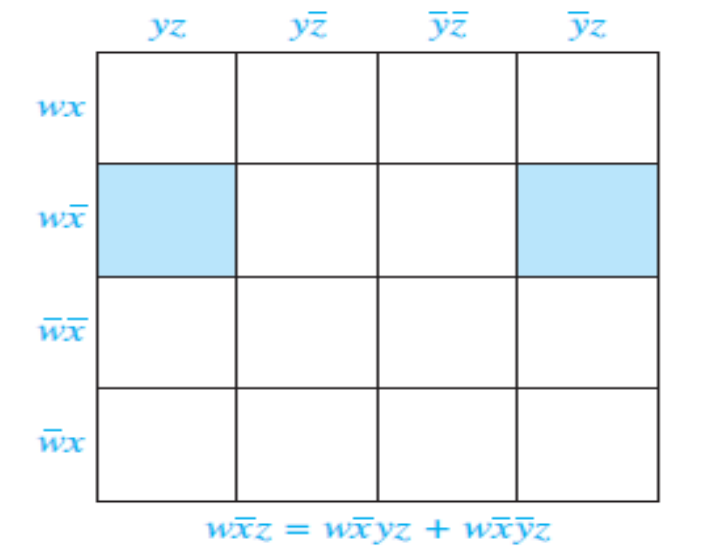
Let $\alpha(x,y,z,w)$ be a sum of product Boolean expression such that minterm consists of three variables x,y,z,w. To define the K-map for $\alpha(x,y,z,w)$, we consider the following table

	yz	$y\bar{z}$	$\bar{y}\bar{z}$	$\bar{y}z$
wx	$wxyz$	$wxy\bar{z}$	$wx\bar{y}\bar{z}$	$wx\bar{y}z$
$w\bar{x}$	$w\bar{x}yz$	$w\bar{x}y\bar{z}$	$w\bar{x}\bar{y}\bar{z}$	$w\bar{x}\bar{y}z$
$\bar{w}\bar{x}$	$\bar{w}\bar{x}yz$	$\bar{w}\bar{x}y\bar{z}$	$\bar{w}\bar{x}\bar{y}\bar{z}$	$\bar{w}\bar{x}\bar{y}z$
$\bar{w}x$	$\bar{w}xyz$	$\bar{w}xy\bar{z}$	$\bar{w}x\bar{y}\bar{z}$	$\bar{w}x\bar{y}z$

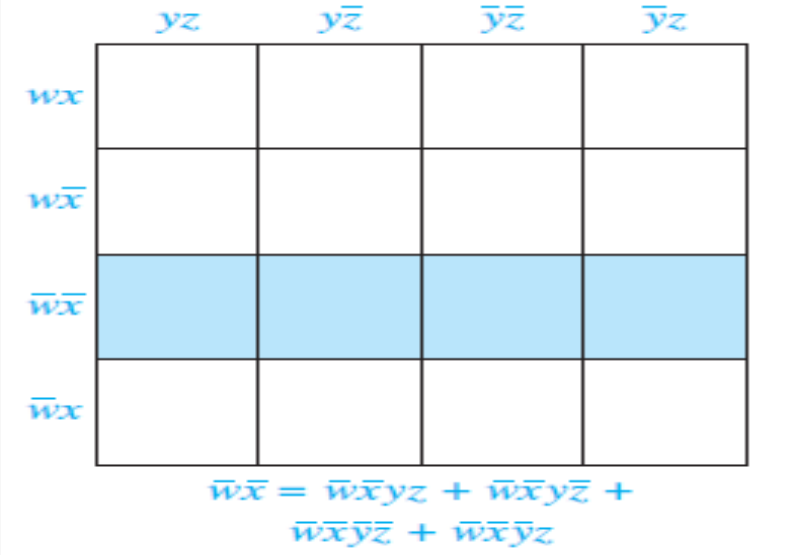
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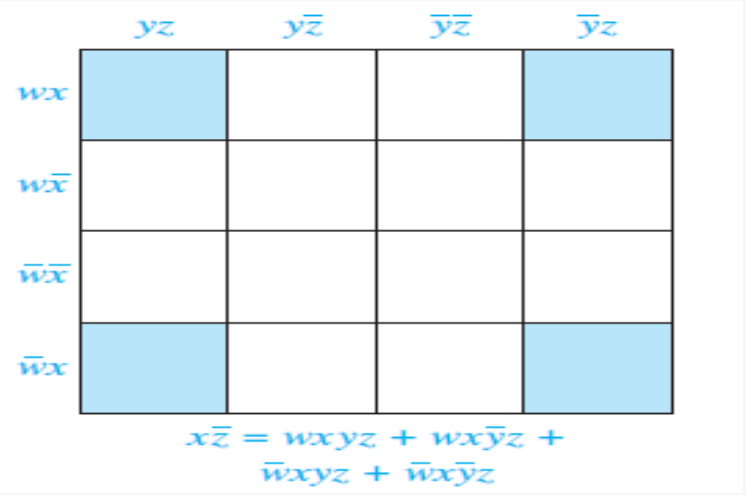
Blocks in K-maps in four variables



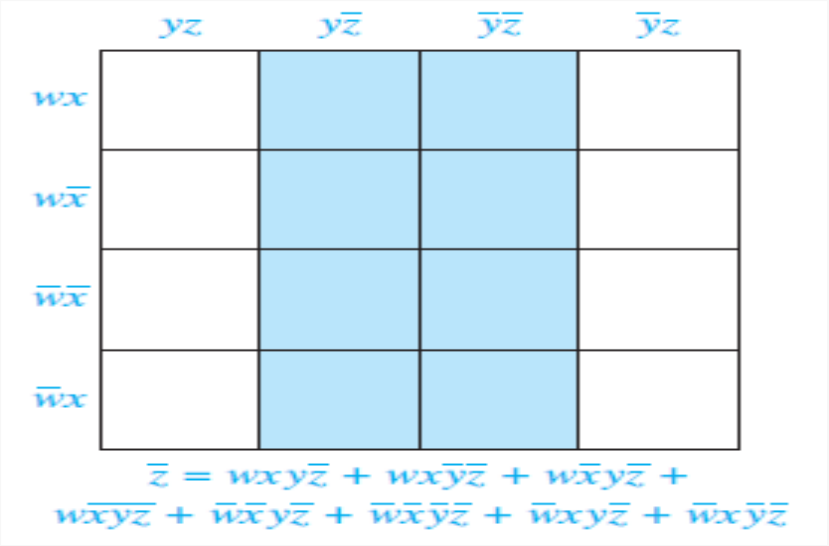
$w\bar{x}z = w\bar{x}yz + w\bar{x}\bar{y}z$



$\bar{w}\bar{x} = \bar{w}\bar{x}yz + \bar{w}\bar{x}y\bar{z} + \bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}\bar{y}z$



$x\bar{z} = wxyz + wx\bar{y}z + \bar{w}xyz + \bar{w}x\bar{y}z$



$\bar{z} = wxy\bar{z} + wx\bar{y}\bar{z} + w\bar{x}y\bar{z} + w\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + \bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}xy\bar{z} + \bar{w}x\bar{y}\bar{z}$

A K-map in four variables

Example :

Use K-maps to simplify these sum-of-products expansions.

(a) $wxyz + wxy\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}y\bar{z} + w\bar{x}\,\bar{y}z + w\bar{x}\,\bar{y}\,\bar{z} + \bar{w}xyz +$ $\bar{w}\,\bar{x}yz + \bar{w}\,\bar{x}y\bar{z}$

Click to add text

(b) $wx\bar{y}\,\bar{z} + w\bar{x}yz + w\bar{x}y\bar{z} + w\bar{x}\,y\,\bar{z} + \bar{w}x\bar{y}\,\bar{z} + \bar{w}\,\bar{x}yz + \bar{w}\,\bar{x}\,y\,\bar{z}$

(c) $wxy\bar{z} + wx\bar{y}\,\bar{z} + w\bar{x}yz + w\bar{x}y\bar{z} + w\bar{x}\,y\,\bar{z} + \bar{w}xyz + \bar{w}xy\bar{z} + \bar{w}x\bar{y}\,\bar{z} +$ $\bar{w}\,\bar{x}yz + \bar{w}\,\bar{x}y\bar{z} + \bar{w}\,x\,\bar{y}\,\bar{z}$

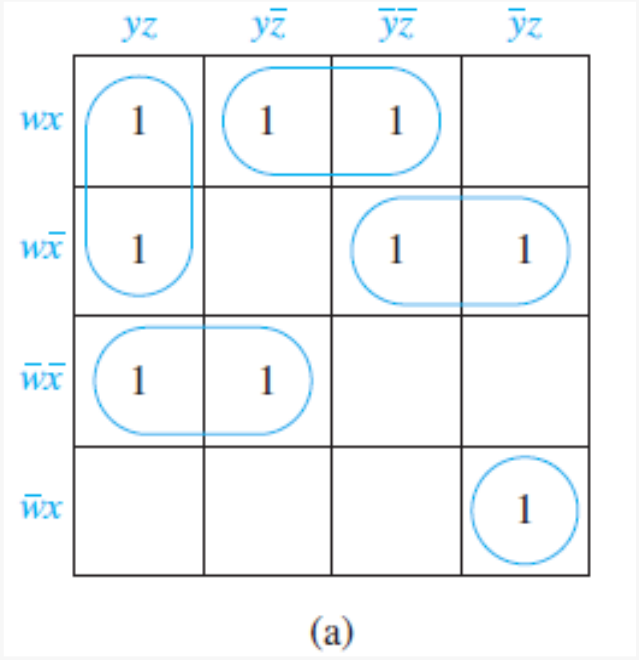
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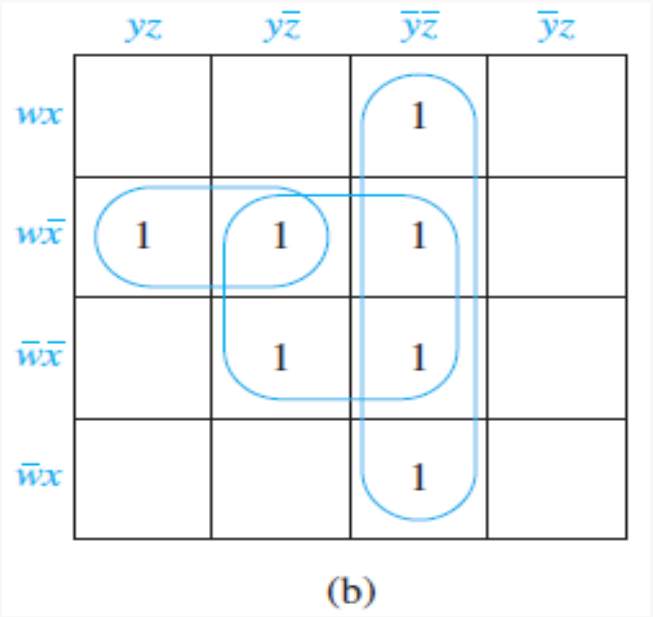
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A K-map in four variables

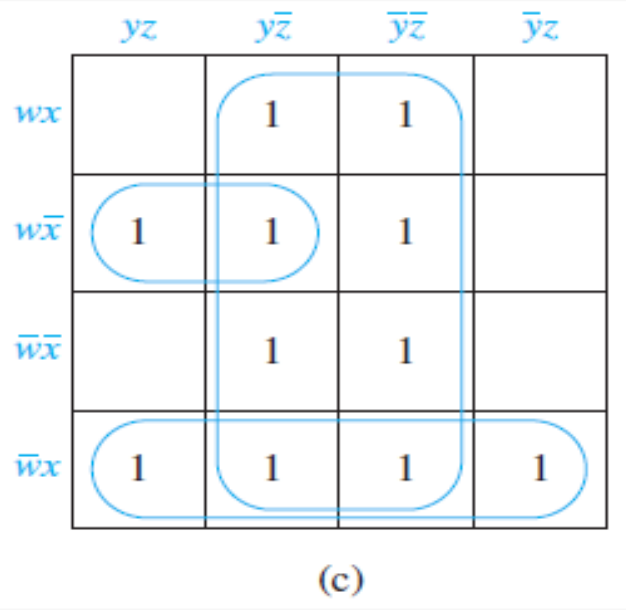
Solution :



$$wyz + wx\bar{z} + w\bar{x}\,\bar{y} + \bar{w}\,\bar{x}y + \bar{w}x\bar{y}z$$



$$\bar{y}\,\bar{z} + w\bar{x}y + \bar{x}\,\bar{z}$$



$$\bar{z} + \bar{w}x + w\bar{x}y$$

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Reference

Kenneth H. Rosen, " Discrete Mathematics and its Applications", 8th edition,2019, McGraw-Hill Education, New York, ISBN 978-1-259-67651-2

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Thank You